Interior Point Methods

CME 307 / MS&E 311 /OIT 676 Optimization

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Newton's method

$$x^+ = x - [\nabla^2 f(x)]^{-1} \nabla f(x)$$

Newton's step converge in one step for quadratic f

A function behaves like a quadratic locally

$$f(x) = f(x^\star) + \frac{1}{2}\langle x - x^\star, \nabla^2 f(x^\star)(x - x^\star)\rangle + \mathcal{O}(\|x - x^\star\|^3)$$

"behaves like" quantified by Lipschitz Hessian

$$\|\nabla^2 f(x) - \nabla^2 f(y)\| \leq H\|x - y\|$$
$$\|\nabla^3 f(x)\| \leq H$$

$$\|x^+ - x^\star\| \leq \frac{H}{\mu}\|x - x^\star\|^2$$

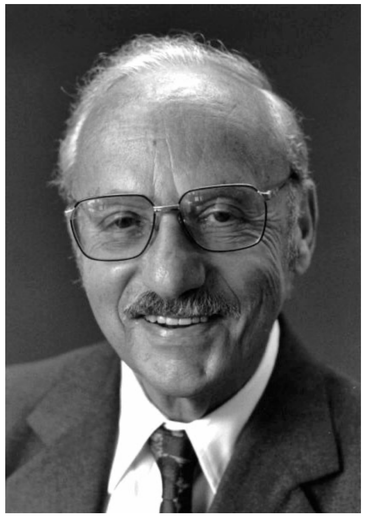
Quasi-Newton amortizes the cost of $[\nabla^2 f(x)]^{-1}$

Review

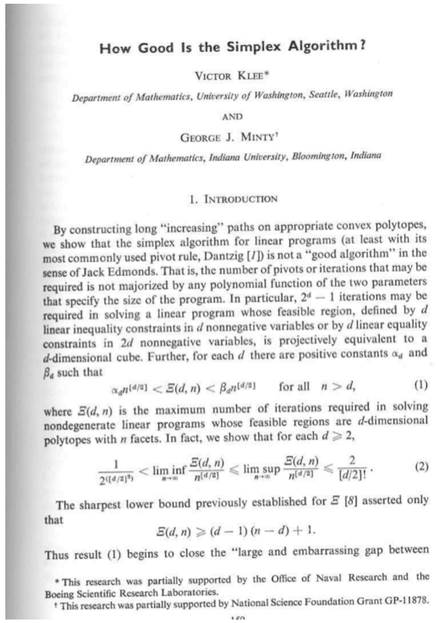
- Newton’s method
- Quasi-Newton method

Today’s goal: understand the insights behind IPM

History of linear programming



$$\min_{x \in \mathbb{R}^n} \quad \langle c, x \rangle$$
$$\text{subject to} \quad Ax = b$$
$$x \geq 0$$



How Good Is the Simplex Algorithm?

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1. INTRODUCTION

By constructing long "increasing" paths on appropriate convex polytopes, we show that the simplex algorithm for linear programs (at least with its most commonly used pivot rule, Dantzig [1]) is not a "good algorithm" in the sense of Jack Edmonds. That is, the number of pivots or iterations that may be required is not majorized by any polynomial function of the two parameters that specify the size of the program. In particular, $2^d - 1$ iterations may be required in solving a linear program whose feasible region, defined by d linear inequality constraints in d nonnegative variables or by d linear equality constraints in $2d$ nonnegative variables, is projectively equivalent to a d-dimensional cube. Further, for each d there are positive constants α_d and β_d such that

$$\alpha_d n^{\lfloor d/2 \rfloor} < \Xi(d, n) < \beta_d n^{\lfloor d/2 \rfloor} \quad \text{for all} \quad n > d, \tag{1}$$

where $\Xi(d, n)$ is the maximum number of iterations required in solving nondegenerate linear programs whose feasible regions are d-dimensional polytopes with n facets. In fact, we show that for each $d \geqslant 2$,

$$\frac{1}{2^{\lfloor d/2 \rfloor}!} < \liminf_{n \to \infty} \frac{\Xi(d, n)}{n^{\lfloor d/2 \rfloor}} \leqslant \limsup_{n \to \infty} \frac{\Xi(d, n)}{n^{\lfloor d/2 \rfloor}} \leqslant \frac{2}{\lfloor d/2 \rfloor!}. \tag{2}$$

The sharpest lower bound previously established for Ξ [8] asserted only that

$$\Xi(d, n) \geqslant (d - 1)(n - d) + 1.$$

Thus result (1) begins to close the "large and embarrassing gap between

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- Simplex method invented by George Dantzig in 1947

  Dramatic practical performance; exponential worst-case in n in theory (1970)

- Stanford MS&E was by then the center of LP and operations research

Search for an efficient polynomial time algorithm





The New York Times

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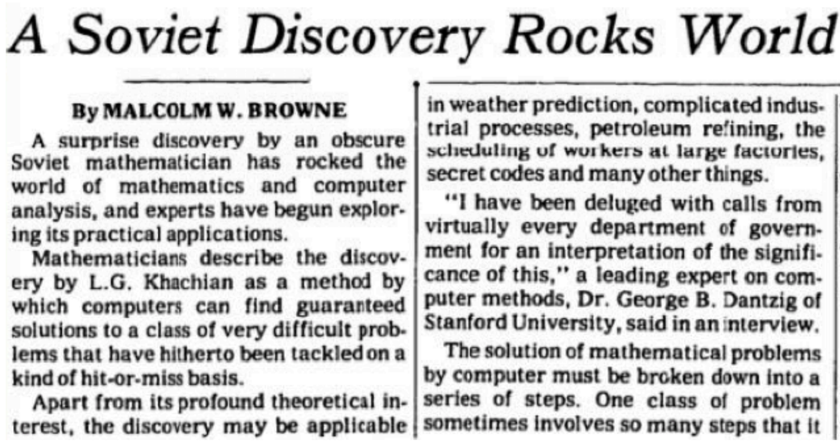
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A Soviet Discovery Rocks World of Mathematics



A Soviet Discovery Rocks World of Mathematics

By MALCOLM W. BROWNE

A surprise discovery by an obscure Soviet mathematician has rocked the world of mathematics and computer analysis, and experts have begun exploring its practical applications.

Mathematicians describe the discovery by L.G. Khachian as a method by which computers can find guaranteed solutions to a class of very difficult problems that have hitherto been tackled on a kind of hit-or-miss basis.

Apart from its profound theoretical interest, the discovery may be applicable in weather prediction, complicated industrial processes, petroleum refining, the scheduling of workers at large factories, secret codes and many other things.

"I have been deluged with calls from virtually every department of government for an interpretation of the significance of this," a leading expert on computer methods, Dr. George B. Dantzig of Stanford University, said in an interview.

The solution of mathematical problems by computer must be broken down into a series of steps. One class of problem sometimes involves so many steps that it could take billions of years to compute.

The Russian discovery offers a way by which the number of steps in a solution can be dramatically reduced. It also offers the mathematician a way of learning quickly whether a problem has a solution or not, without having to complete the entire immense computation that may be required.

According to the American journal Sci-

Continued on Page A20, Column 3

ONLY $10.00 A MONTH!!! 24 Hr. Phone Answering Service. Totally New Concept!!! Incredible!!! 279-3870—ADVT.

- In 1979, Khachiyan found that LP can be solved in $\mathcal{O}(n^4 L)$ iterations with the ellipsoid method
- The ellipsoid method is theoretically elegant but has poor performance

Then the interior point method comes

Dikin's affine scaling

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \langle c, x \rangle \\ \text{subject to} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

Starting from feasible $x > 0$, find d such that $x + d$ remains feasible

$$\begin{aligned} \min_{d} \quad & \langle c, d \rangle \\ \text{subject to} \quad & A(x + d) = b \\ & \|X^{-1}d\| \leq 1 \end{aligned}$$

- When $x > 0$, the update ask how far to go with respect to each coordinate
- $\{d : \|X^{-1}d\| \leq 1\}$ is called Dikin's ellipsoid

Dikin's affine scaling

$$\min_{x \in \mathbb{R}^n} \quad \langle c, x \rangle \qquad \text{subject to} \quad Ax = b, \quad x \geq 0$$

Starting from feasible $x > 0$, find d such that $x + d$ remains feasible

$$d' \leftarrow X^{-1} d$$

$$\min_{d} \quad \langle c, d \rangle \qquad \text{subject to} \quad Ad = 0 \qquad \|X^{-1} d\| \leq 1$$

$$\min_{d'} \quad \langle Xc, d' \rangle \qquad \text{subject to} \quad AXd' = 0 \qquad \|d'\| \leq 1$$

Projected gradient

$$d = -\frac{\overbrace{XP_{AX}Xc}}{\|P_{AX}Xc\|}$$

$$P_{AX} = I - XA^\top (AX^2A^\top)^{-1} AX$$

Dikin's affine scaling

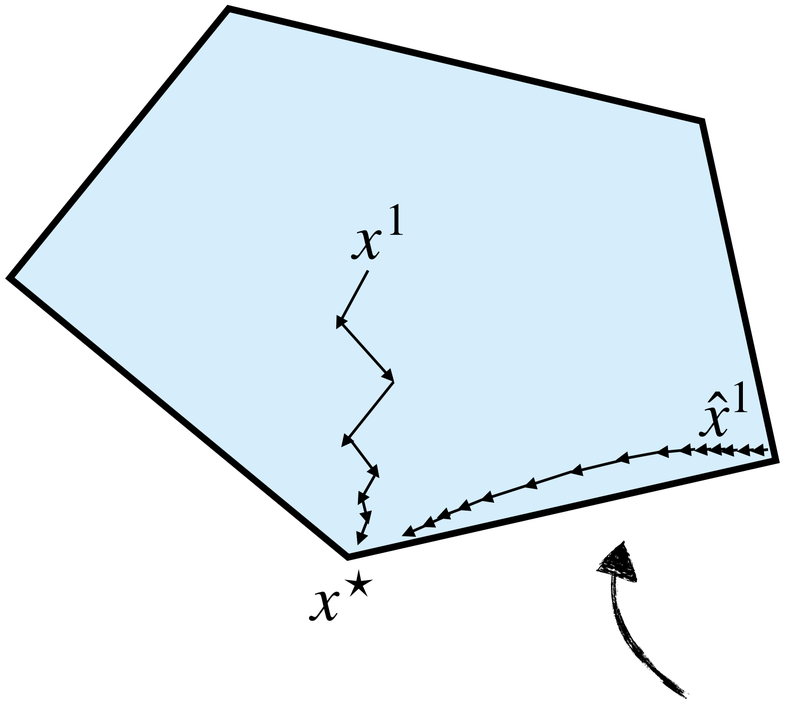
Initialize $x^1 > 0$

$$d_k = -\frac{X^k P_{AX^k} X^k c}{\|P_{AX^k} X^k c\|} \qquad x^{k+1} = x^k + t d_k, \quad t < 1$$

- x^k is feasible all the time
- Good performance on a few LP problems
- Slow on some problems

Failure of affine scaling IPM [demo]

$$d_k = -\frac{X^k P_{AX^k} X^k c}{\|P_{AX^k} X^k c\|}$$

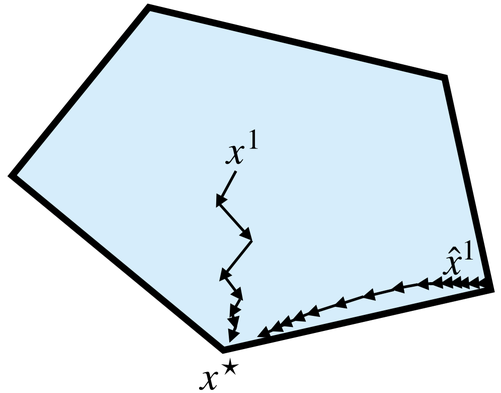


Size of d_k is proportional to x^k

Affine scaling can be trapped by a bad initial point

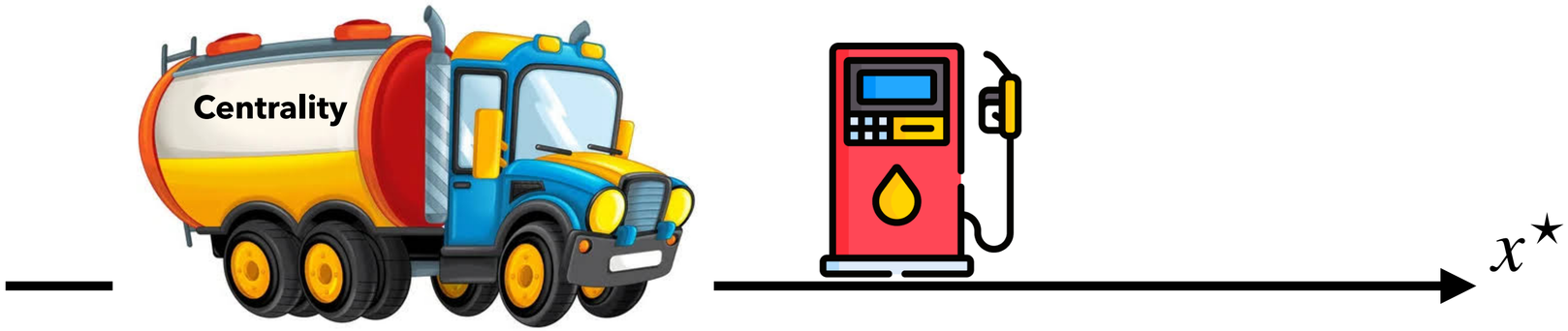
What makes these two points different?

Centrality



Affine scaling works well when the point is at the center

Centrality provides a chance to recover from biased direction



Idea: consume centrality sparingly when approaching optimality

+ restore centrality when it's insufficient

Analytic center

Given a polytope $\{x : Ax = b, x \geq 0\}$, its analytic center is the minimizer of

$$\max_x \overbrace{\sum_{i=1}^{n} x_i \log x_i}^{\phi(x)} \quad \text{subject to} \quad Ax = b \qquad x \geq 0$$

The non-centrality can be measured by the gradient in a Dikin step

$$\nabla \phi(x) = -X^{-1}\mathbf{1}$$

$$P_{AX} X \nabla \phi(x) = -P_{AX}\mathbf{1}$$

Balancing optimality and centrality

$$\phi(x) = -\sum_{i=1}^{n} \log x_i$$

At point x, we have

(Sub-) optimality measured by

$$P_{AX} X c$$

(Non-) centrality measured by

$$P_{AX} X \nabla \phi(x)$$

Combines these two directions with parameter $\mu > 0$

$$d = -XP_{AX}X\left(\frac{1}{\mu}c + \nabla\phi(x)\right) = -XP_{AX}\left(\frac{Xc}{\mu} - \mathbf{1}\right)$$

Logarithmicc Barrier

$$\phi(x) = -\sum_{i=1}^{n} \log x_i$$

Projected gradient

$$d = -XP_{AX}X\Big(\frac{1}{\mu}c + \nabla\phi(x)\Big)$$

$$\frac{1}{\mu}\nabla\Big[\frac{1}{\mu}\langle c, x\rangle + \mu\phi(x)\Big]$$

Gradient norm:

$$\delta_\mu(x) := \|P_{AX}\Big(\frac{Xc}{\mu} - \mathbf{1}\Big)\|$$

$$\min_{Ax=b,x\geq 0} \langle c, x\rangle + \mu\phi(x)$$

Central path

$$\delta_\mu(x) := \|P_{AX}\left(\frac{Xc}{\mu} - \mathbf{1}\right)\|$$

- As $\mu \to 0$, $\delta_\mu(x) \approx \frac{1}{\mu}\|P_{AX}Xc\|$ and measures suboptimality

$$\langle c, x\rangle - \langle c, x^\star\rangle \leq 2\mu n \text{ for } \delta_\mu(x) \leq 1$$

- When $\delta_\mu(x) < 1$, x is well-centered with respect to parameter μ and

$$\delta_\mu(x + d) \leq [\delta_\mu(x)]^2$$

- The curve $\{x(\mu), \mu > 0 : Ax = b, \delta_\mu(x) = 0\}$ is called the central path

$$x(\mu) = \arg \min_{Ax=b, x\geq 0} \langle c, x\rangle + \mu\phi(x)$$

- The region $N_\mu = \{x : Ax = b, \delta_\mu(x) \leq 1\}$ is called the central path neighborhood at μ

Finding $x \in N_\mu$ for sufficiently small μ

Path-following [demo]

For $\delta_\mu(x) \le 1$:

$$\delta_\mu(x+d) \le [\delta_\mu(x)]^2$$

$$\langle c, x\rangle - \langle c, x^\star\rangle \le 2\mu n$$

Finding $x \in N_\mu$ for sufficiently small μ

Strategy 1. Fix small μ and optimize aggressive for fixed μ

Strategy 2. Start from a large μ and progressively reduce it

Homotopy path-following method

Path-following

$$x^+ = x - XP_{AX}\left(\frac{Xc}{\mu} - \mathbf{1}\right)$$

For $\delta_\mu(x) \leq \frac{1}{2} < 1$

$$\langle c, x\rangle - \langle c, x^\star\rangle \leq 2\mu n \qquad \delta_\mu(x^+) \leq [\delta_\mu(x)]^2 \leq \frac{1}{4} < \frac{1}{2}$$

After one step decreases δ_μ , we can update $\mu \leftarrow \mu^+$ so that

- $\delta_{\mu^+}(x^+) \leq \frac{1}{2}$
- Then $\langle c, x^+\rangle - \langle c, x^\star\rangle \leq 2\mu^+ n$

Convergence of IPM is determined by the reduction in μ

Polynomiality of IPM

Initialize $\mu^1 = 1, \quad \delta_{\mu^1}(x^1) \le \frac{1}{2}$

$$x^{k+1} = x^k - X^k P_{AX^k}\left(\frac{X^k c}{\mu^k} - \mathbf{1}\right)$$

$$\mu^{k+1} \leftarrow (1-\theta)\mu^k$$

$$\delta_{\mu^k}(x^{k+1}) \le [\delta_{\mu^k}(x^k)]^2 \le \frac{1}{4}$$

so that $\quad \delta_{\mu^{k+1}}(x^{k+1}) \le \frac{1}{2}$

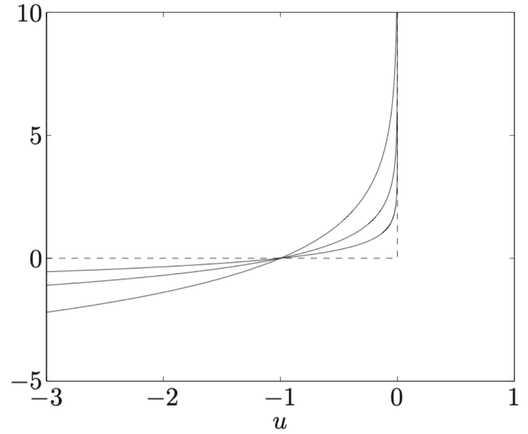
Theorem. It's feasible to take $\theta = \frac{1}{8\sqrt{n}}$.

$$\mu^{k+1} \le (1 - \frac{1}{8\sqrt{n}})\mu^k$$

$$\langle c, x^{K+1}\rangle - \langle c, x^\star\rangle \le 2\mu^{K+1} n \le 2(1-\frac{1}{8\sqrt{n}})^K n$$

$$\Rightarrow \mathcal{O}(\sqrt{n}\log(\frac{n}{\varepsilon})) \text{ iterations for } \varepsilon \text{ accuracy}$$

Barrier penalty and KKT form



How solver developers often derive IPM

$$\begin{aligned} \min_{x\in\mathbb{R}^n} \quad & \langle c, x\rangle \\ \text{subject to} \quad & Ax = b \\ & x \geq 0 \end{aligned} \quad \Rightarrow \quad \begin{aligned} \min_{x\in\mathbb{R}^n} \quad & \langle c, x\rangle - \mu \sum_{i=1}^{n} \log x_i \\ \text{subject to} \quad & Ax = b \end{aligned}$$

Barrier $-\log x$ serves as an approximation of $\mathbb{I}\{x \geq 0\}$

Driving parameter $\mu \to 0$ progressively improves approximation

KKT form and Newton step

$$\min_{x \in \mathbb{R}^n} \quad \langle c, x \rangle - \mu \sum_{i=1}^{n} \log x_i \qquad \max_{y,s} \quad \langle b, y \rangle + \mu \sum_{i=1}^{n} \log s_i$$

subject to $Ax = b$ subject to $A^\top y + s = c$

The KKT condition is given by

$$\begin{aligned} Ax &= b \\ A^\top y + s &= c \\ s - \mu X^{-1}\mathbf{1} &= 0 \end{aligned} \qquad \begin{aligned} A(x + d_x) &= b \\ A^\top (y + d_y) + (s + d_s) &= c \\ s + d_s - \mu X^{-1}\mathbf{1} + \mu X^{-2} d_x &= 0 \end{aligned}$$

👍 Allows (x, y, s) that do not satisfy the equality conditions

Primal-dual and dual IPM

$$A(x + d_x) = b \qquad A^\top(y + d_y) + s + d_s = c$$

Primal IPM

$$s - \mu X^{-1}\mathbf{1} = 0$$

$$s + d_s - \mu X^{-1}\mathbf{1} + \mu X^{-2} d_x = 0$$

Primal-dual IPM

$$XS\mathbf{1} - \mu\mathbf{1} = 0$$

$$Xd_s + Sd_x + XS\mathbf{1} - \mu\mathbf{1} = 0$$

Dual IPM

$$x - \mu S^{-1}\mathbf{1} = 0$$

$$x + d_x - \mu S^{-1}\mathbf{1} + \mu S^{-2} d_s = 0$$



Implementation

The key computation in IPM is solving the linear system

$$
\begin{aligned}
A(x + d_x) &= b \\
A^\top(y + d_y) + (s + d_s) &= c \\
s + d_s + \mu X^{-2}(x + d_x) &= 0
\end{aligned}
$$

$$
\begin{pmatrix} 0 & A^\top & I \\ A & 0 & 0 \\ \mu X^{-2} & 0 & I \end{pmatrix}
\begin{pmatrix} d_x \\ d_y \\ d_s \end{pmatrix} =
\begin{pmatrix} r_d \\ r_p \\ r_c \end{pmatrix}.
$$

Augmented system (quasi-definite)

$$
\begin{pmatrix} -\mu X^{-2} & A^\top \\ A & 0 \end{pmatrix}
\begin{pmatrix} d_x \\ d_y \end{pmatrix} =
\begin{pmatrix} h \\ r_p \end{pmatrix}.
$$

Normal equation (positive-definite)

$$
AX^2A^\top d_y = w
$$

More implementation

The key computation in IPM is solving the linear system

Almost no one solves

$$\begin{pmatrix} 0 & A^\top & I \\ A & 0 & 0 \\ \mu X^{-2} & 0 & I \end{pmatrix} \begin{pmatrix} d_x \\ d_y \\ d_s \end{pmatrix} = \begin{pmatrix} r_d \\ r_p \\ r_c \end{pmatrix}$$

Some solve

$$\begin{pmatrix} -X^{-2} & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} d_x \\ d_y \end{pmatrix} = \begin{pmatrix} h \\ r_p \end{pmatrix}$$

Many solve

$$AX^2A^\top d_y = w$$

$$ADA^\top d = r$$

- Analyze the sparsity pattern of $ADA^\top$
- Factorize $PADA^\top P^\top = LL^\top$
- Solve for d
- Iterative refinement (for higher accuracy)

Practical performance

Iter	Primal.Obj	Dual.Obj	Compl	Primal.Inf	Dual.Inf	Time
0	+1.73051376e+04	+3.14331798e+02	5.99e+04	1.56e+03	1.96e+01	0.49s
1	+5.24131224e+03	-3.25290528e+02	1.10e+04	2.15e+02	3.08e+00	0.50s
2	+9.51739796e+02	+4.84745156e+01	1.03e+03	1.63e+00	1.23e-01	0.50s
3	+5.13471553e+02	+1.45588908e+02	3.67e+02	6.77e-01	1.36e-03	0.50s
4	+4.55820481e+02	+2.40142970e+02	2.14e+02	4.38e-01	7.27e-04	0.50s
5	+3.24215163e+02	+2.77338335e+02	4.65e+01	8.85e-02	1.28e-04	0.50s
6	+2.95732755e+02	+2.80911795e+02	1.47e+01	2.58e-02	2.67e-05	0.50s
7	+2.90139601e+02	+2.81887980e+02	8.19e+00	1.47e-02	9.03e-06	0.50s
8	+2.85000111e+02	+2.82532985e+02	2.45e+00	4.16e-03	2.37e-06	0.50s
9	+2.83156890e+02	+2.82838865e+02	3.17e-01	3.42e-04	6.84e-07	0.50s
10	+2.82978845e+02	+2.82936906e+02	4.19e-02	1.67e-05	1.46e-07	0.50s
11	+2.82960676e+02	+2.82960153e+02	5.23e-04	9.05e-08	2.30e-09	0.50s
12	+2.82960474e+02	+2.82960474e+02	3.56e-08	1.40e-11	2.66e-14	0.50s

IPM is arguably the most efficient algorithm for LPs (along with simplex)

- Primal-dual IPM is widely implemented in commercial and open-source solvers
- Also the most efficient for medium-scale conic problems
- ≤ 50 iterations for most problems
- Local superlinear convergence

Self-concordance

$$\delta_\mu(x+d) < [\delta_\mu(x)]^2 \qquad \text{for } \delta_\mu(x) \le 1$$

Have we seen this kind of behavior before? $\|x^+ - x^\star\| \le \frac{H}{\mu}\|x - x^\star\|^2$

$f(x) = \frac{1}{\mu}\langle c, x\rangle - \sum_i \log x_i$ does not have a Lipschitz Hessian $\frac{\mathrm{d}^3}{\mathrm{d}x^3}[-\log x] = -2x^{-3}$

It turns out the following condition suffices for Newton's method

Self-concordance: $|f'''(x)| \le 2f''(x)^{3/2}$

Self-concordant barrier functions

$$|f'''(x)| \leq 2f''(x)^{3/2}$$

A useful application for self-concordance is constructing penalty functions for cones

- Logarithmic barrier $-\log x$ for $x \geq 0$
- Log-determinant barrier $-\log\det X$ for $X \succeq 0$
- Barrier for second-order cone $-\log(t^2 - \|x\|^2)$ for $(x, t) \succeq_{\text{SOC}} 0$

Summary

- Balance between optimality and centrality
- Homotopy path-following analysis
- Implementation & practical performance

History

~1950
Log barrier/penalty

1967
Dikin's affine scaling

1984
Karmarkar's projective transformation
Polynomial-time LP

1986-1990
Path-following IPM
Primal-dual IPM

1994
Self-dual embedding

1997
Self-concordance
Conic programming

2014
Improved complexity